```c
#include <stdio.h>

int main()
{
    int n, i, current_time = 0, completed_count = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int pid[n], at[n], bt[n], pr[n];
    int ct[n], tat[n], wt[n], rt[n], st[n];
    int completed[n];

    // Input
    for(i = 0; i < n; i++)
    {
        printf("\nEnter Process ID: ");
        scanf("%d", &pid[i]);

        printf("Enter Arrival Time: ");
        scanf("%d", &at[i]);

        printf("Enter Burst Time: ");
        scanf("%d", &bt[i]);

        printf("Enter Priority: ");
```

```c
    scanf("%d", &pr[i]);

    completed[i] = 0;
}

while(completed_count < n)
{
    int highest_priority = 9999;
    int selected = -1;

    for(i = 0; i < n; i++)
    {
        if(at[i] <= current_time && completed[i] == 0)
        {
            if(pr[i] < highest_priority)
            {
                highest_priority = pr[i];
                selected = i;
            }
        }
    }

    if(selected == -1)
    {
```

```
        current_time++;
    }
    else
    {
        st[selected] = current_time;

        ct[selected] = st[selected] + bt[selected];

        tat[selected] = ct[selected] - at[selected];

        wt[selected] = tat[selected] - bt[selected];

        rt[selected] = st[selected] - at[selected];

        current_time = ct[selected];

        completed[selected] = 1;
        completed_count++;
    }
}

printf("\nPID\tAT\tBT\tPR\tCT\tTAT\tWT\tRT\n");

for(i = 0; i < n; i++)
{
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
```

```
        pid[i], at[i], bt[i], pr[i], ct[i], tat[i], wt[i], rt[i]);
    }

    return 0;
}
```

"C:\Users\Admin\Desktop\

```
Enter Process ID: 2
Enter Arrival Time: 2
Enter Burst Time: 2
Enter Priority: 3

Enter Process ID: 3
Enter Arrival Time: 3
Enter Burst Time: 5
Enter Priority: 2

Enter Process ID: 4
Enter Arrival Time: 4
Enter Burst Time: 4
Enter Priority: 4

Enter Process ID: 5
Enter Arrival Time: 6
Enter Burst Time: 1
Enter Priority: 1

PID     AT      BT      PR      CT      TAT     WT      RT
1       0       3       5       3       3       0       0
2       2       2       3       11      9       7       7
3       3       5       2       8       5       0       0
4       4       4       4       15      11      7       7
5       6       1       1       9       3       2       2

Process returned 0 (0x0)   execution time : 87.742 s
Press any key to continue.
```

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